VINUNIVERSITY

Test duration: 1 hour and 30 minutes

Name: **Student ID:**

Midterm Practice Test 2 – Solutions – Fall 2026

COMP3040/5030 Computer Vision Unofficial practice test, prepared by Lê Tăng Phú Quý

Instructions:

- Academic honesty is required in all work you submit to be graded. You should **NOT** copy or share your solutions with other students to avoid plagiarism issues.
- For short answer questions, please make your answers as **concise** as possible.
- This is a closed-book exam and NO Internet access is allowed. You are allowed 1 page of **hand-written** note (both sides).
- Drafting paper and pen/pencil are allowed. We will provide the drafting paper to you.
- Late submission without an approved extension will not be graded at all.
- You must **RETURN** this hard copy of the midterm exam at the end of the exam period.
- During the exam, you are **NOT** allowed to leave the room or/and use the restroom.
- You are required to **NOT** use your mobile phone in the exam room. Please put it on the table or in your bag.
- Good luck!

About the exam: There are **05** problems with **100 points** total, and some have multiple parts. Please check through the pages before you start.

Problem	Content	Point (Undergraduates)	Type
1	Fundamentals of Computer Vision	20	short answer
2	Image Filtering	25	long answer
3	Feature Detection and Matching	25	long answer
4	Fitting and Image Transformations	25	long answer
5	System Design	05	long answer
Total		**100**	

Problem 1. Fundamentals of Computer Vision

Give a short answer for each of the following question:

1.1. Distinguish image classification, object detection and semantic segmentation by the **output** of each task. (5 pts)

Solution. Classification: one label per image. Detection: a set of bounding boxes + a label each. Semantic segmentation: a label for **every pixel**.

1.2. A 6.4 mm wide sensor has a 90° horizontal FOV. What is the focal length? If the focal length doubles, what is the new FOV? (5 pts)

Solution. $f = \frac{d}{2\tan(45°)} = 3.2$ mm. With $f = 6.4$ mm: FOV $= 2\arctan(3.2/6.4) = 2 \cdot 26.57° \approx 53.1°$ (not $45°$).

1.3. What is **aliasing**? How do you avoid it when shrinking an image $\times 2$? (5 pts)

Solution. Sampling at $f_s < 2f_{\max}$ folds high frequencies into false low ones (moiré, jaggies). Avoid it by low-passing first (Gaussian $\sigma \approx 1$ or 2×2 averaging) and only then dropping rows/columns — a Gaussian pyramid, `INTER_AREA`.

1.4. Explain the **aperture** trade-off of a pinhole camera and the role of a lens. (5 pts)

Solution. Small hole: few rays per image point ⇒ sharp but dark (too small ⇒ diffraction). Large hole: bright but each scene point becomes a blur spot. A lens gathers many rays and focuses them to a point ⇒ bright and sharp, but only for objects near one distance (depth of field).

Problem 2. Image Filtering

2.1. (a) Show that the 2-D Gaussian kernel is **separable**. (b) For a 7×7 kernel, how many multiplications per pixel does direct 2-D filtering cost, and separable filtering? (c) Blurring with $\sigma = 6$ then $\sigma = 8$ equals a single blur with which σ? (12 pts)

Solution. (a) $e^{-(x^2+y^2)/2\sigma^2} = e^{-x^2/2\sigma^2} e^{-y^2/2\sigma^2} \Rightarrow G = \mathbf{g}\mathbf{g}^\top$: filter rows then columns. (b) 49 vs 14. (c) Variances add: $\sqrt{36 + 64} = 10$.

2.2. Signal $f = [3, 1, 4, 1, 5, 9, 2]$, kernel $k = [1, 2, 3]$. Compute (a) the **correlation** and (b) the **convolution**, 'valid' mode. (c) Which one is commutative? Why does it matter when chaining filters? (13 pts)

Solution. (a) $o[i] = f[i] + 2f[i+1] + 3f[i+2]$: $[17, 12, 21, 38, 29]$.

(b) Flip the kernel to $[3, 2, 1]$: $[15, 12, 19, 22, 35]$.

(c) Convolution is commutative **and** associative: $(A * B) * I = A * (B * I) \Rightarrow$ several filters merge into one kernel (e.g. derivative of Gaussian). Correlation is not commutative.

Problem 3. Feature Detection and Matching

3.1. (a) Why does the DoG $L(k\sigma) - L(\sigma)$ approximate the scale-normalized Laplacian of

Gaussian? (b) A bright disc of radius 6 px responds most strongly at which σ? (c) With $s = 3$ intervals per octave, what is k, and how many Gaussian and DoG images per octave? (10 pts)

Solution. (a) The heat equation $\partial G/\partial\sigma = \sigma\nabla^2 G \Rightarrow G(k\sigma) - G(\sigma) \approx (k-1)\sigma^2\nabla^2 G$. (b) $\sigma^* = r/\sqrt{2} \approx 4.24$. (c) $k = 2^{1/3} \approx 1.26$; $s+3 = 6$ Gaussian images, $s+2 = 5$ DoGs.

3.2. A SIFT candidate has the spatial Hessian $H = \begin{bmatrix} 12 & 2 \\ 2 & 1 \end{bmatrix}$. With $r = 10$, keep or reject? Explain geometrically. (8 pts)

Solution. $\mathrm{tr}^2 / \det = 169/8 = 21.1 > (11)^2/10 = 12.1 \Rightarrow$ **reject**: one principal curvature dominates — the point lies along an **edge**, so its position along the edge is unstable.

3.3. Descriptor $\mathbf{a}$ has three nearest neighbours in the other image at distances $0.21, 0.24, 0.65$. With a ratio test of 0.8, is its best match kept? Why is the ratio test better than an absolute threshold? (7 pts)

Solution. $0.21/0.24 = 0.875 > 0.8 \Rightarrow$ **reject** (ambiguous: two near-equal candidates). Distinctiveness varies across descriptors; the ratio test asks whether the best match **stands out** from the next confusable one — it removes ~90% of false matches while losing <5% of correct ones (Lowe).

Problem 4. Fitting and Image Transformations

4.1. Three image points $(2,0), (1,1), (0,2)$. (a) Write the line through them in normal form $\rho = x\cos\theta + y\sin\theta$ with $\theta \in [-90°, 90°)$. (b) What curve does each point produce in (θ, ρ) space? (c) Why use (ρ, θ) instead of (m, b)? (12 pts)

Solution. (a) $x + y = 2 \Rightarrow$ divide by $\sqrt{2}$: $\theta = 45°$, $\rho = \sqrt{2} \approx 1.41$. (b) A point $(x_0, y_0) \mapsto$ the sinusoid $\rho = x_0\cos\theta + y_0\sin\theta$: $2\cos\theta$, $\cos\theta + \sin\theta$, $2\sin\theta$ — all three meet at $(45°, \sqrt{2})$. (c) m, b are unbounded (a vertical line has $m = \infty$); (ρ, θ) lives in the bounded domain $[-d, d] \times [-90°, 90°)$.

4.2. An affine map sends $(0,0) \to (1,2)$, $(1,0) \to (3,3)$, $(0,1) \to (0,4)$. (a) Find its 2×3 matrix. (b) The image of $(2,2)$? (c) Is it a similarity (rotation + uniform scale + translation)? If so, find the angle and scale. (13 pts)

Solution. (a) $\mathbf{b} = (1,2)$; column 1 $= (3,3) - (1,2) = (2,1)$; column 2 $= (0,4) - (1,2) = (-1,2)$ $\Rightarrow \begin{bmatrix} 2 & -1 & 1 \\ 1 & 2 & 2 \end{bmatrix}$. (b) $(4 - 2 + 1,\ 2 + 4 + 2) = (3, 8)$. (c) Yes: the form $\begin{bmatrix} a & -c \\ c & a \end{bmatrix}$ with $a = 2, c = 1$ $\Rightarrow$ scale $\sqrt{5} \approx 2.24$, angle $\arctan(1/2) \approx 26.6°$.

Problem 5. System Design

5. Design a phone **panorama mode**: the user sweeps the phone and takes 5 overlapping photos. Describe the automatic stitching algorithm and name at least two failure cases with mitigations. (5 pts)

Solution. SIFT on every image $\to$ match adjacent pairs (NN + ratio 0.75) $\to$ RANSAC homography (4 points, a few-px threshold), refit on inliers $\to$ pick the middle image as reference,

chain homographies $\rightarrow$ canvas from the warped corners plus a translation $T \rightarrow$ inverse warp with bilinear sampling $\rightarrow$ blending (feathering/multi-band) + exposure compensation. Failures: **parallax** when the hand translates (near scene) $\Rightarrow$ instruct to rotate about the camera, prefer distant overlap; **moving objects** (people) $\Rightarrow$ ghosting, choose optimal seams; **textureless regions** (sky) $\Rightarrow$ few matches, use gyroscope priors; accumulated error $\Rightarrow$ middle reference / bundle adjustment.